

Week 3 — Factors, Multiples, HCF/LCM & Number Patterns — Answer Key

Q1. B) 7 — $84 = 7 \times 12$, so 7 is a factor. (5, 9, 11 don't divide 84 exactly.)

Q2. A) 39 — $39 = 13 \times 3$. (42, 45, 48 are not multiples of 13.)

Q3. C) 12 — $48 = 2^4 \times 3$; $60 = 2^2 \times 3 \times 5$. Common factors: $2^2 \times 3 = 12$.

Q4. B) 36 — Multiples of 12: 12, 24, 36... Multiples of 18: 18, 36... First common one = 36.

Q5. C) 59 — 59 has no factors besides 1 and itself. ($51=3 \times 17$, $57=3 \times 19$, $63=9 \times 7$.)

Q6. C) 9 — Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36 — that's 9 factors.

Q7. B) 36 — LCM of 4 and 9 (they share no common factor) = $4 \times 9 = 36$.

Q8. D) 115 — Multiples of 11: ...77, 88, 99, 110, 121... 115 isn't among them.

Q9. C) 79 — Each term is "double, then add 1": $4 \times 2 + 1 = 9$, $9 \times 2 + 1 = 19$, $19 \times 2 + 1 = 39$, $39 \times 2 + 1 = 79$.

Q10. C) 42 — These follow $n \times (n+1)$: $1 \times 2 = 2$, $2 \times 3 = 6$, $3 \times 4 = 12$, $4 \times 5 = 20$, $5 \times 6 = 30$, $6 \times 7 = 42$.

Q11. C) 49 — Square numbers: $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2 = 1, 4, 9, 16, 25, 36, 49$.

Q12. C) 18 — $36 = 2^2 \times 3^2$, $54 = 2 \times 3^3$, $90 = 2 \times 3^2 \times 5$. Common factors: $2 \times 3^2 = 18$.

Q13. C) 8 — $96 \div 12 = 8$ baskets.

Q14. B) 10:12 a.m. — They ring together again after $\text{LCM}(18, 24) = 72$ minutes = 1 h 12 min; 9:00 a.m. + 1 h 12 min = 10:12 a.m.

Q15. A) 15 — 15 is a factor of 60 ($60 \div 15 = 4$), a multiple of 5, and greater than 10. (8 isn't even a multiple of 5; 14 and 25 aren't factors of 60.)

Scoring: 15/15 — excellent, move to Week 4. 13–14/15 — solid, review the missed ones. Below 13/15 — go back over factors vs. multiples and HCF vs. LCM — they're easy to mix up and everything later builds on telling them apart quickly.
